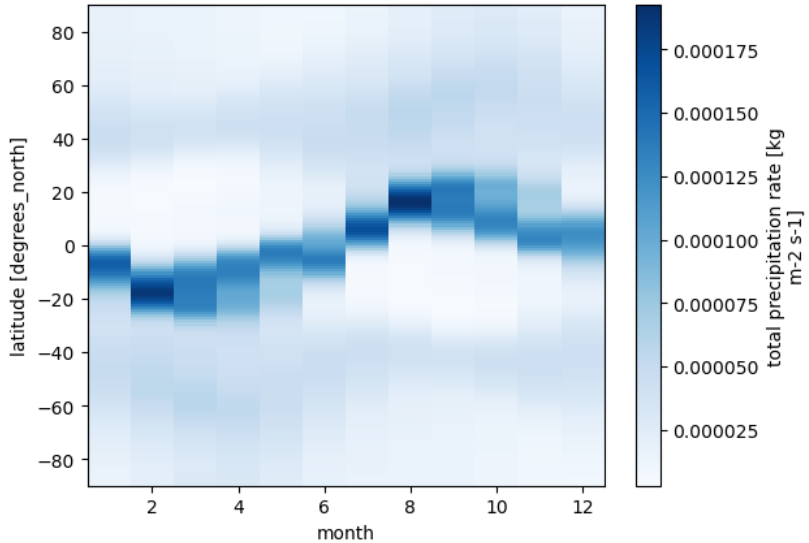


Ole Merkes

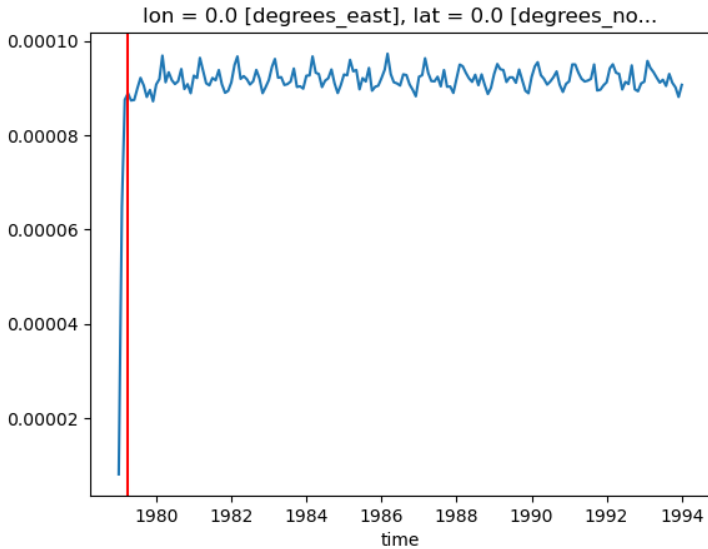
University of Hamburg

09.12.2025

SML depth



Spin up



Scaling of the HC

Angular momentum conserving

$$\Phi_H \sim \left(\frac{gH\Delta_h}{\Omega^2 R_E^2} \right)^{1/2} \quad (1)$$

$$\psi_{\max} \sim \frac{1}{\tau} \frac{g^{3/2} (H\Delta_h)^{5/2}}{\Omega^3 R_E^2 \Delta_v} \quad (2)$$

Baroclinic instability

$$\theta_C \sim \left(\frac{NH}{2\Omega a} \right)^{1/2} \quad (3)$$

4x CO₂

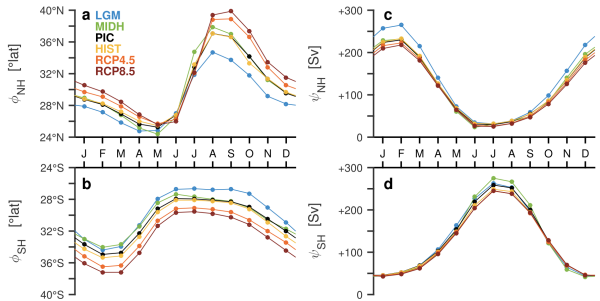
- global warming: tropopause height and stability increase while meridional temperature gradient in the subtropical atmosphere decreases

Research gap?

- D'Agostino, R., P. Lionello, O. Adam, and T. Schneider (2017), Factors controlling Hadley circulation changes from the Last Glacial Maximum to the end of the 21st century, *Geophys. Res. Lett.*, 44, 8585–8591, doi:10.1002/2017GL074533.

Research gap?

- PMIP3 and CMIP5 data from the Last Glacial Maximum (LGM), Mid-Holocene (MIDH), Pre-Industrial Control (PIC), Historical (HIST), and Representative Concentration Pathways 4.5 and 8.5 experiments



Research gap?

Multimodel ensemble mean stream function difference between LGM, MIDH, and RCP8.5 and relative to PIC

